

Management of patients with neurological diseases

Dr . Bardies shaban

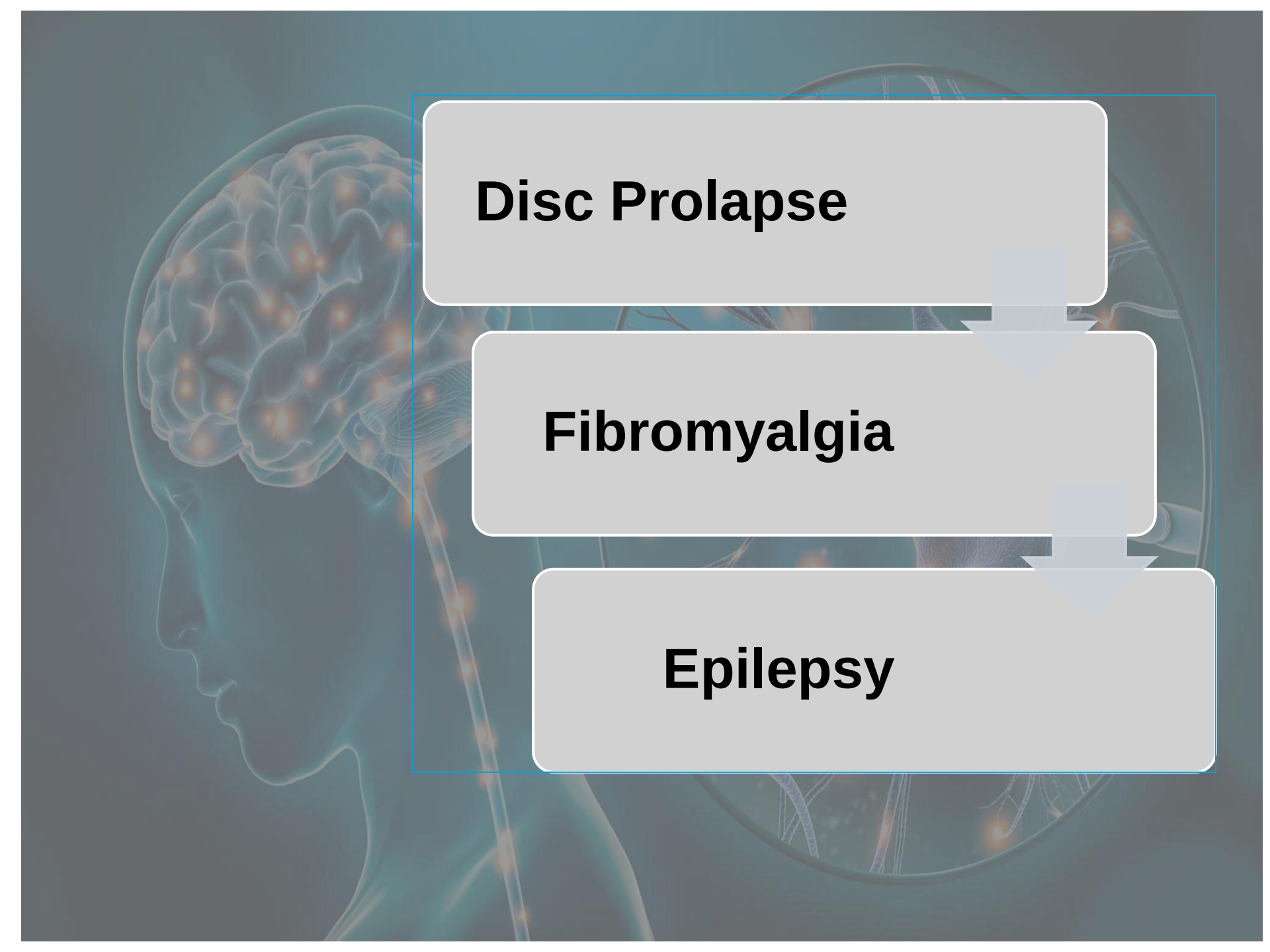




الحديث الجانبي يشتت الانتباه



من فضلك اغلق الهاتف

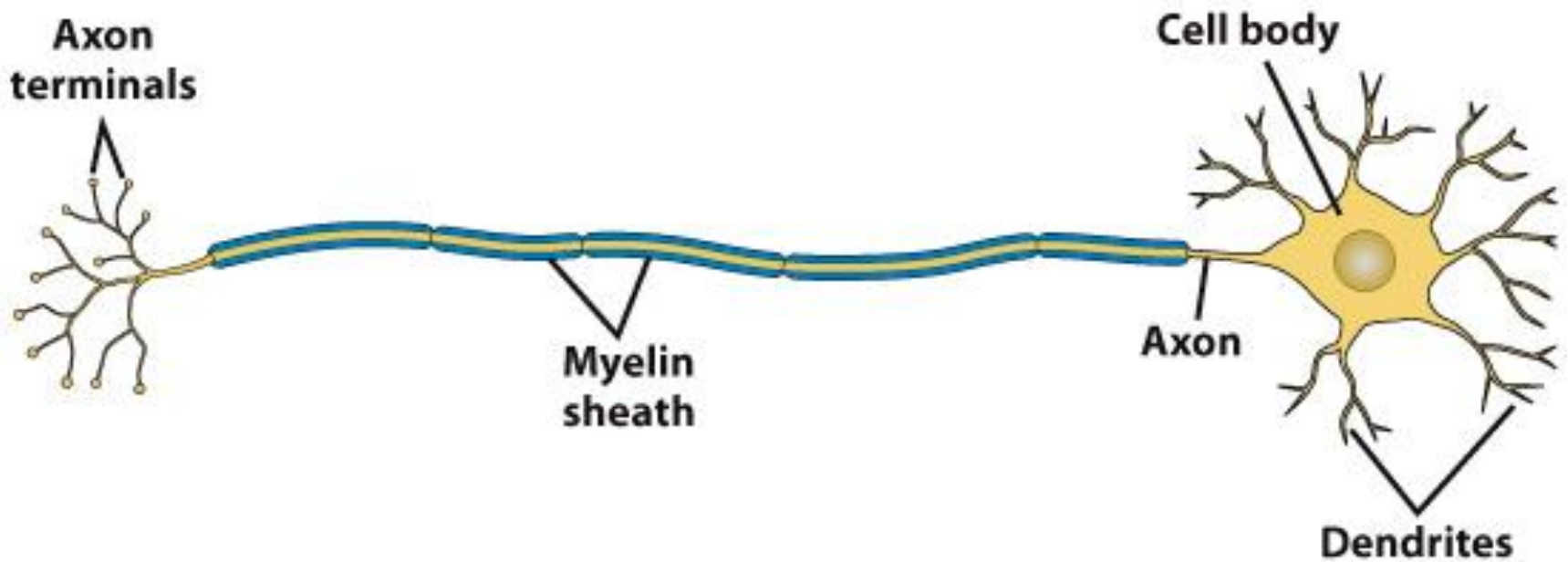


Disc Prolapse

Fibromyalgia

Epilepsy

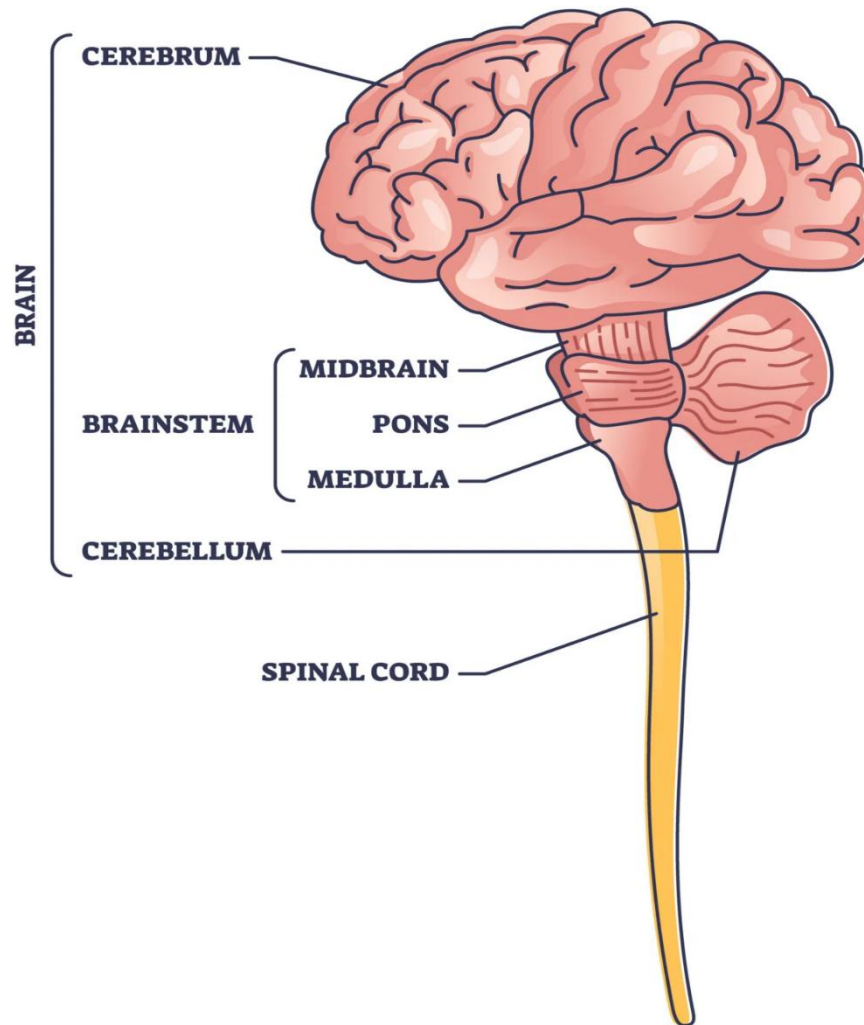
Introduction



The functional unit of the nervous system is the neuron, which consists of a **cell body**, **dendrites**, and an **axon**.

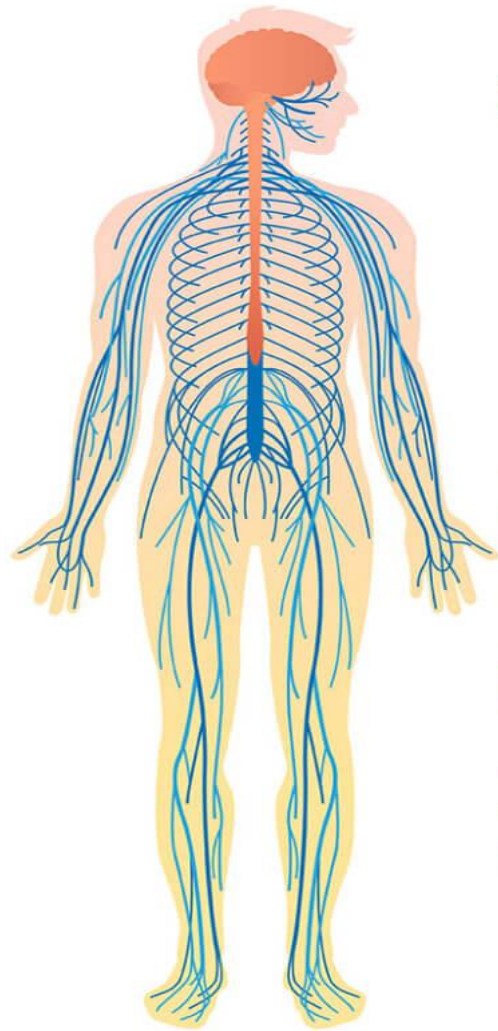
The nervous system consists of the **central nervous system** (CNS) and the **peripheral nervous system** (PNS).

CENTRAL NERVOUS SYSTEM

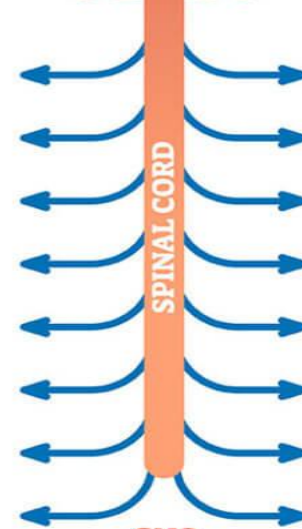


Central
nervous
system

PERIPHERAL NERVOUS SYSTEM



BRAIN



Peripheral Nervous System

Peripheral Nervous System

CNS
Central
nervous
System

CENTRAL NERVOUS SYSTEM

VS

PERIPHERAL NERVOUS SYSTEM

✧ Brain and spinal cord

✧ Controls thinking and emotions

✧ Processes and interprets info

✧ Handles reflexes (via spinal cord)

✧ Nerves outside the CNS

✧ Controls movement and sensation

✧ Sends signals to/from the CNS

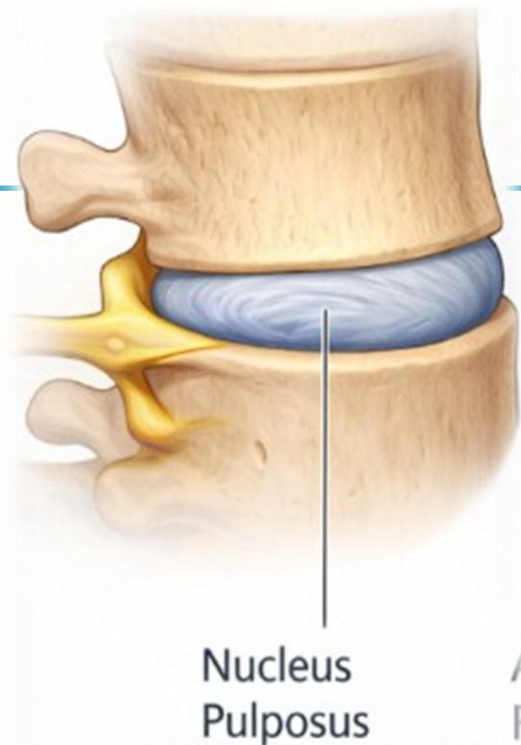
✧ Includes somatic and autonomic systems

1- Disc Prolapse

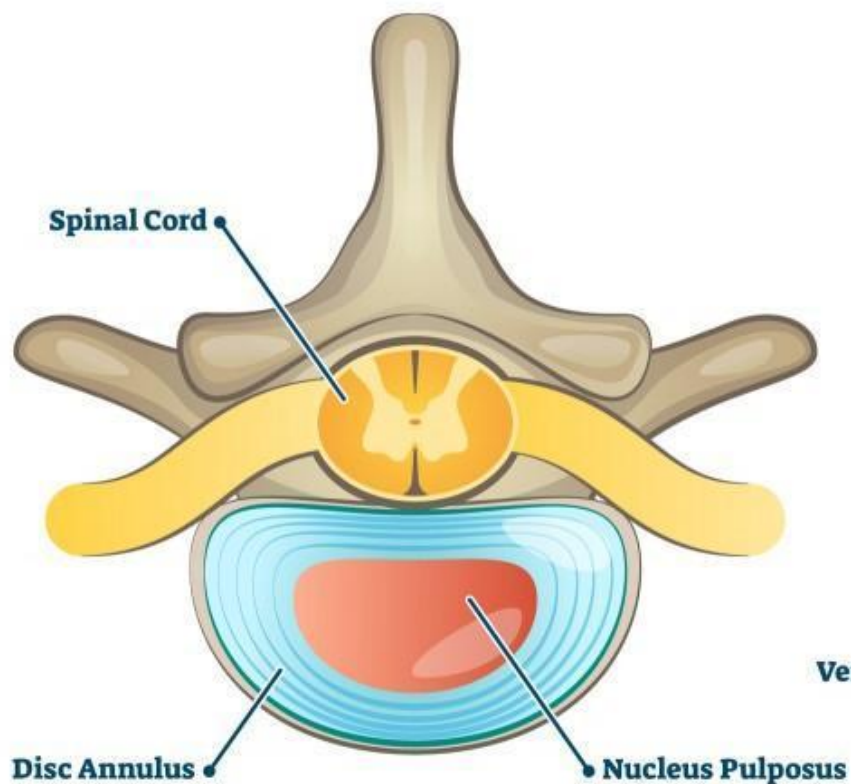
- Definition
- Causes
- Manifestations
- Diagnosis
- Complications
- Managements

Definition

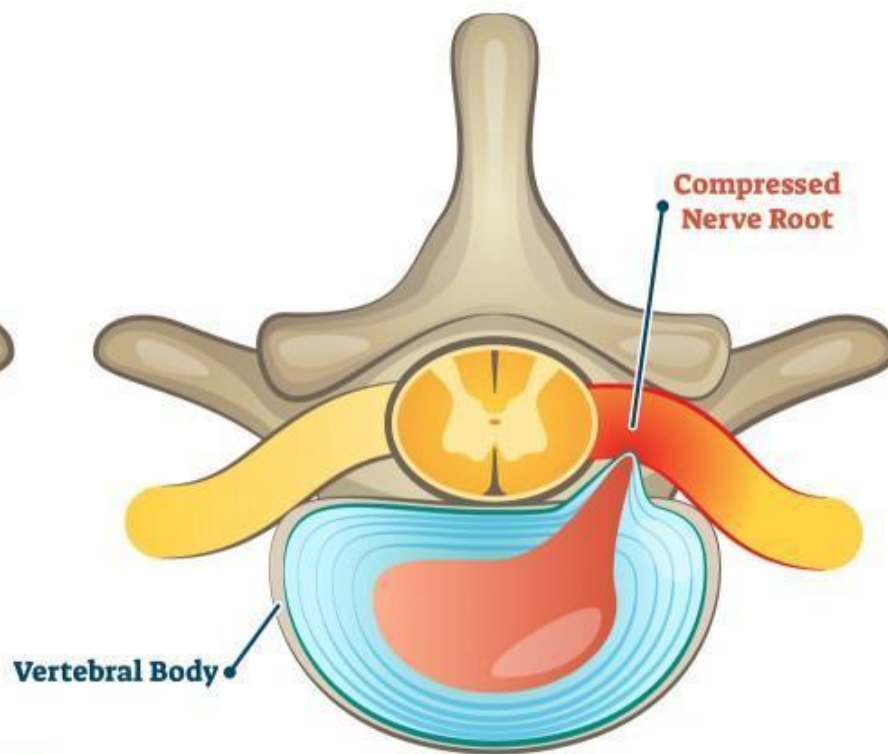
A disc prolapse occurs when the (nucleus pulposus); soft inner core of a spinal disc **bulges out** and **press on** surrounding nerves or spinal cord.



HERNIATED DISC



NORMAL DISC



HERNIATED DISK

Normal Intervertebral
Disc



Nucleus
Pulposus

Disc Bulge



Annulus
Fibrosus

Repetitive use or trauma
can cause inner gel to
press against the outer layer,
creating a bulge.

MOST COMMON LOCATIONS

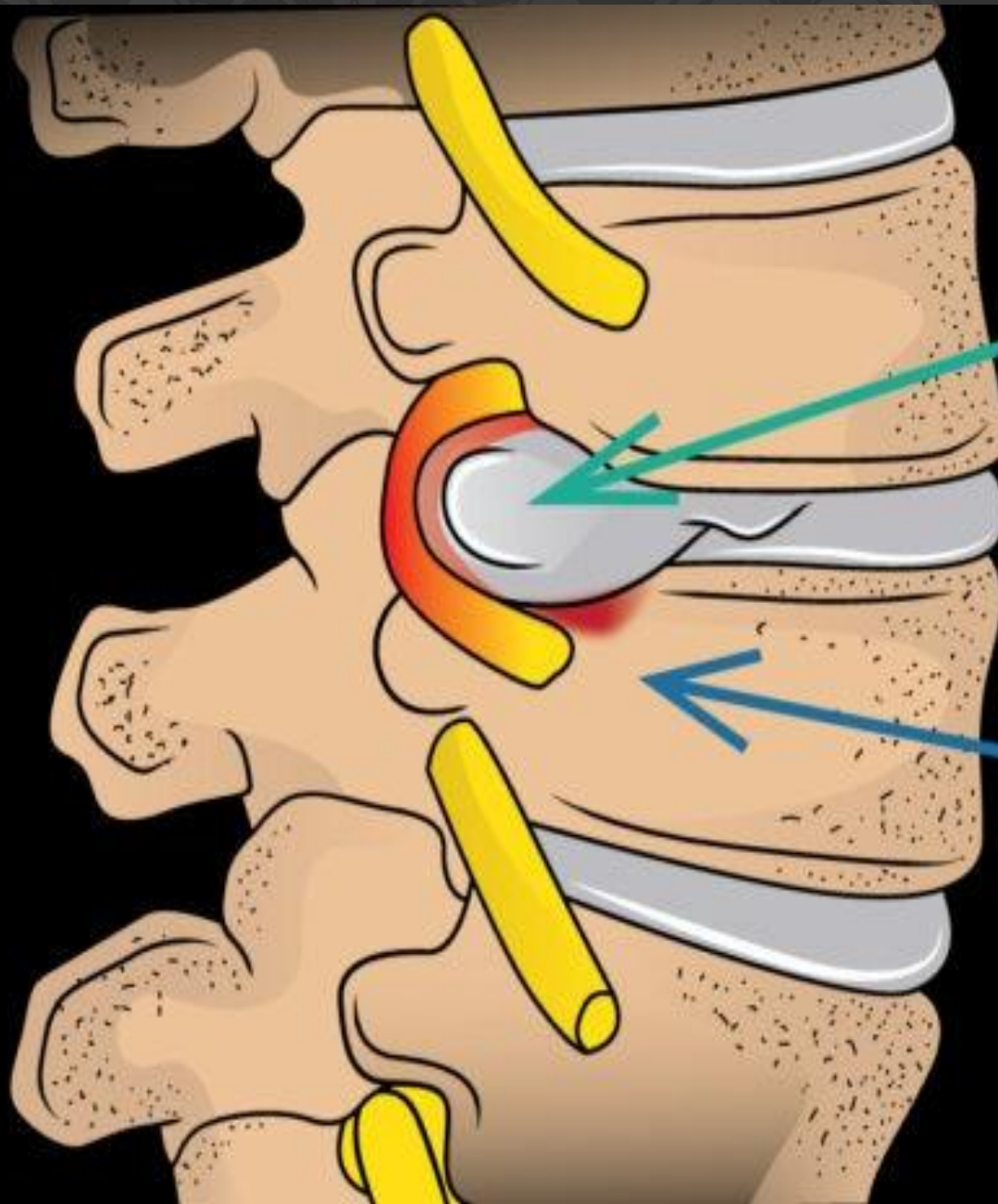


Lumbar
(Lower Back)



Cervical
(Neck)

The most common place for a disc bulge to occur is in
the lower back, but it can also happen in the neck.



Lumbar Disc
Herniation

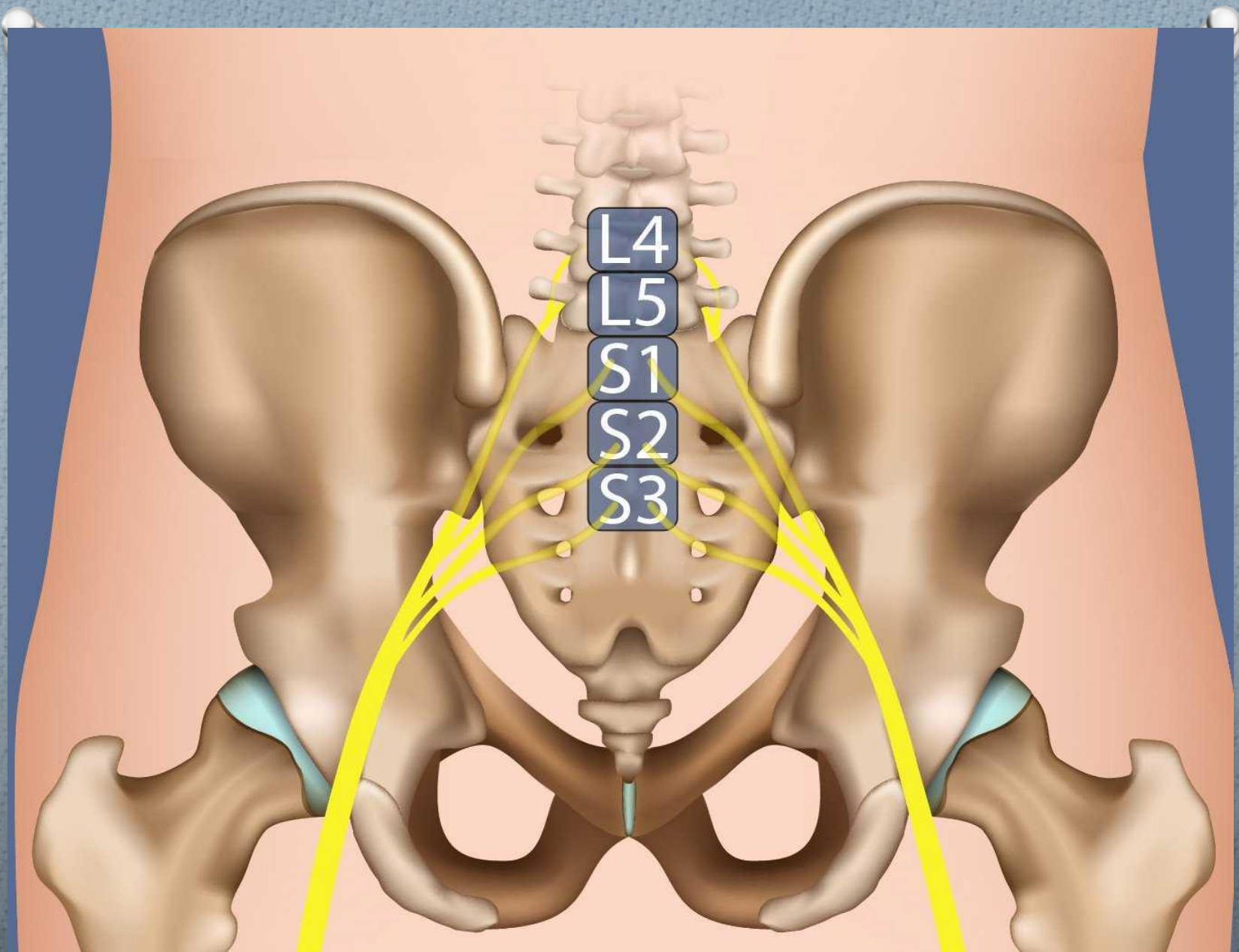
Nerve Root
Compression

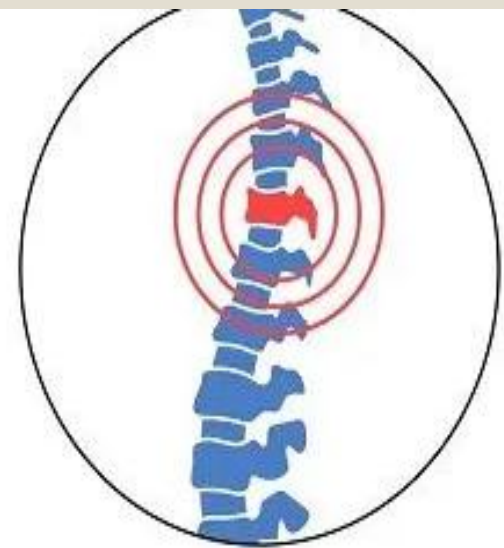
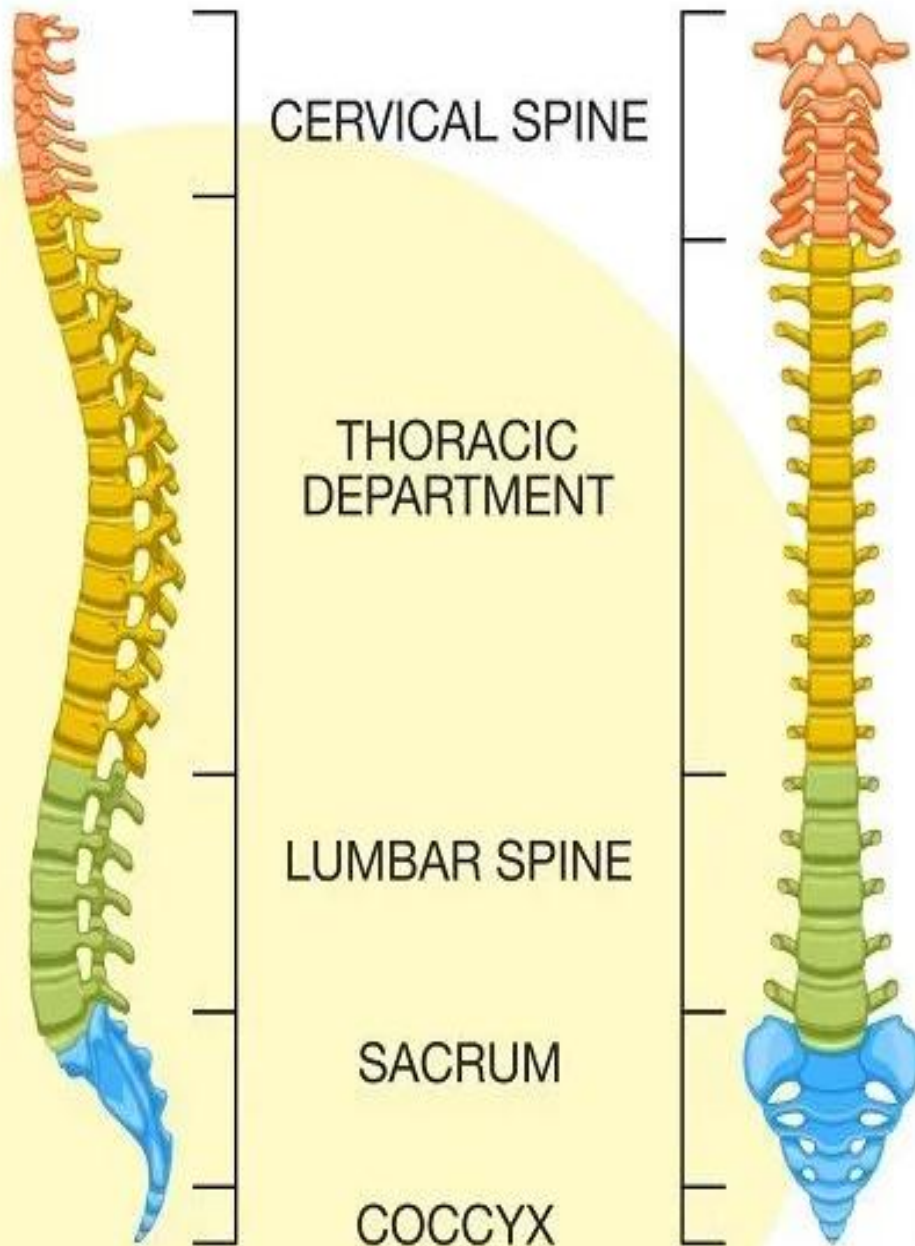
o It most commonly affects the lumbar spine (L4–L5, S1) but can also occur in the cervical or thoracic spine

o Cervical disc prolapse

o Thoracic disc prolapse

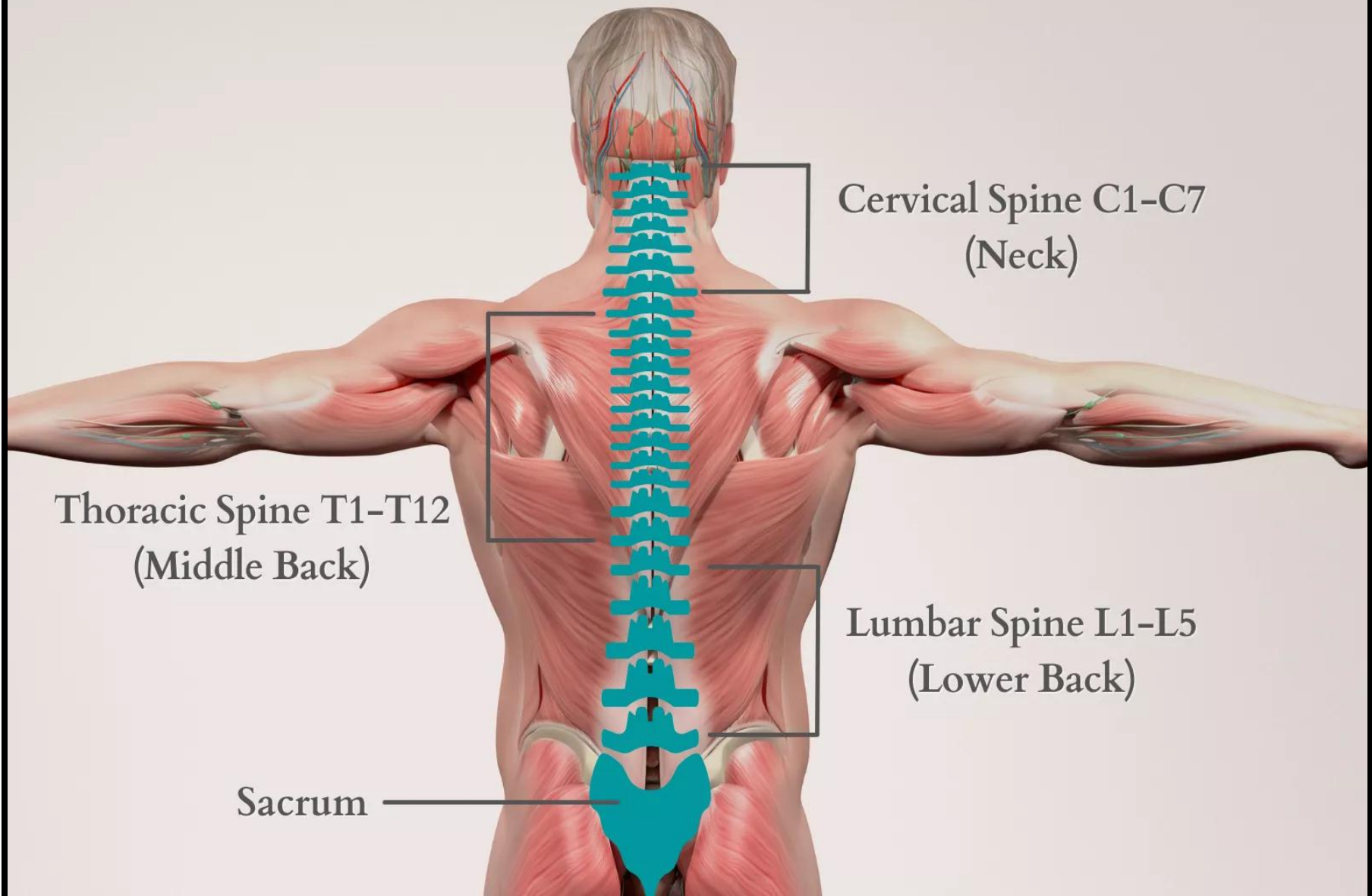
o lumbar disc prolapse





Based on the location, slipped discs are majorly classified into three types, namely

- Cervical disc herniation
- Thoracic disc herniation
- Lumbar disc herniation



Cervical Spine C1-C7
(Neck)

Thoracic Spine T1-T12
(Middle Back)

Lumbar Spine L1-L5
(Lower Back)

Sacrum

Causes

- **Degenerative disc disease** ; age-related wear and tear
- **Trauma or injury** ;e.g., lifting heavy objects improperly
- **Repetitive strain or twisting motions**
- **Obesity** ; increased spinal load
- **Poor posture**
- **Sedentary lifestyle**
- **Genetic predisposition**

Causes of Disc Prolapse



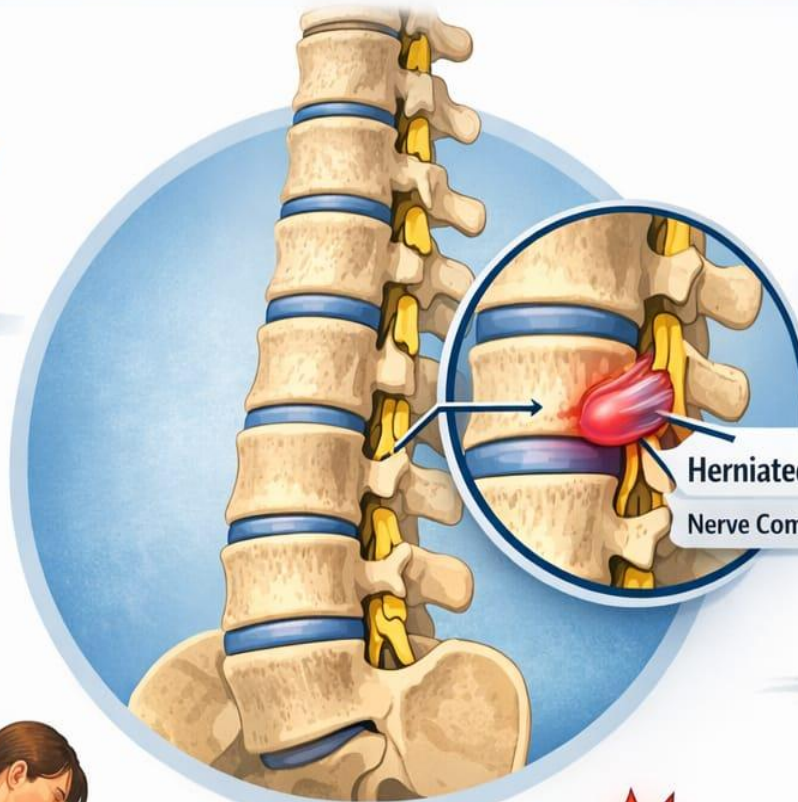
Aging & Degeneration



Heavy Lifting & Poor Posture



Obesity



**Herniated Disc
Nerve Compression**



Injury & Trauma



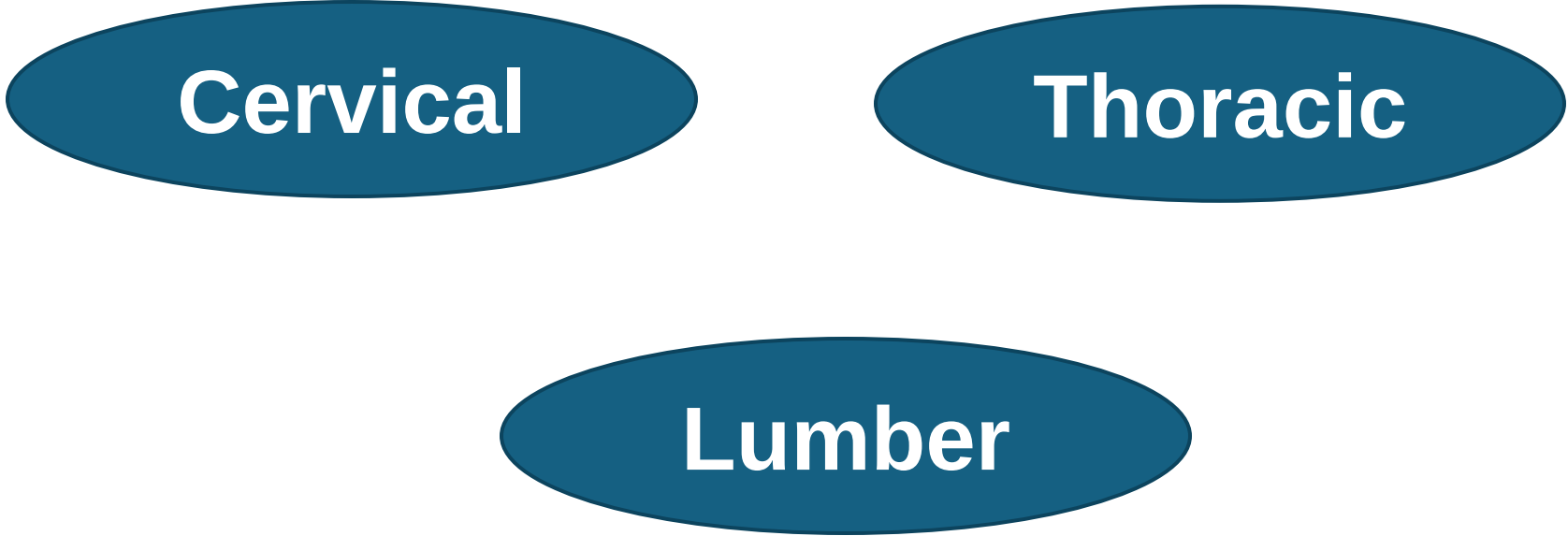
Sedentary Lifestyle



Smoking & Alcohol

Signs & Symptoms

Symptoms depend on the location and degree of nerve compression:



Cervical

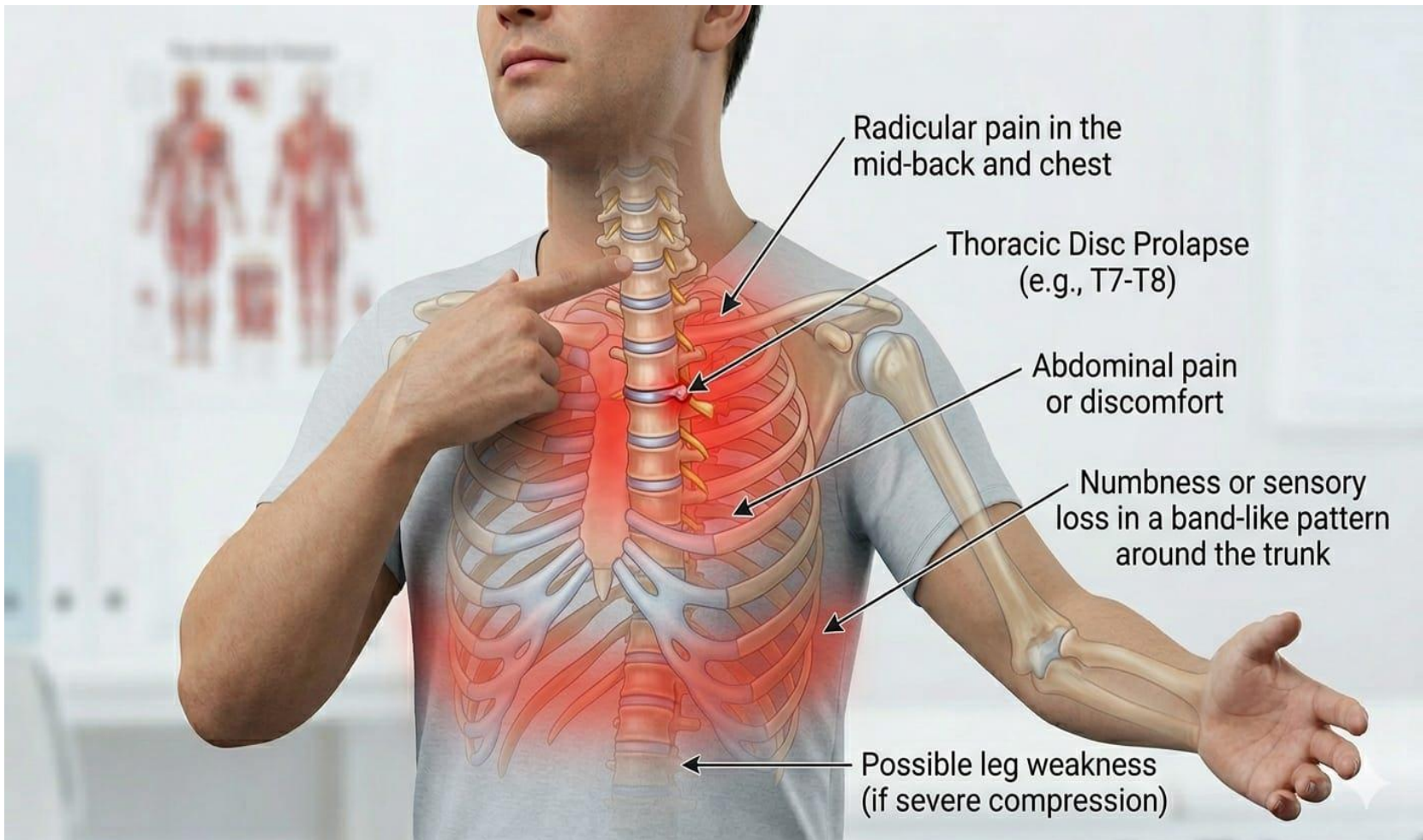
Thoracic

Lumber

Symptoms of **cervical** disc prolapse



Symptoms of **thoracic** disc prolapse



Symptoms of Lumbar Disc Prolapse



Low Back Pain



Numbness & Tingling



Numbness & Tingling



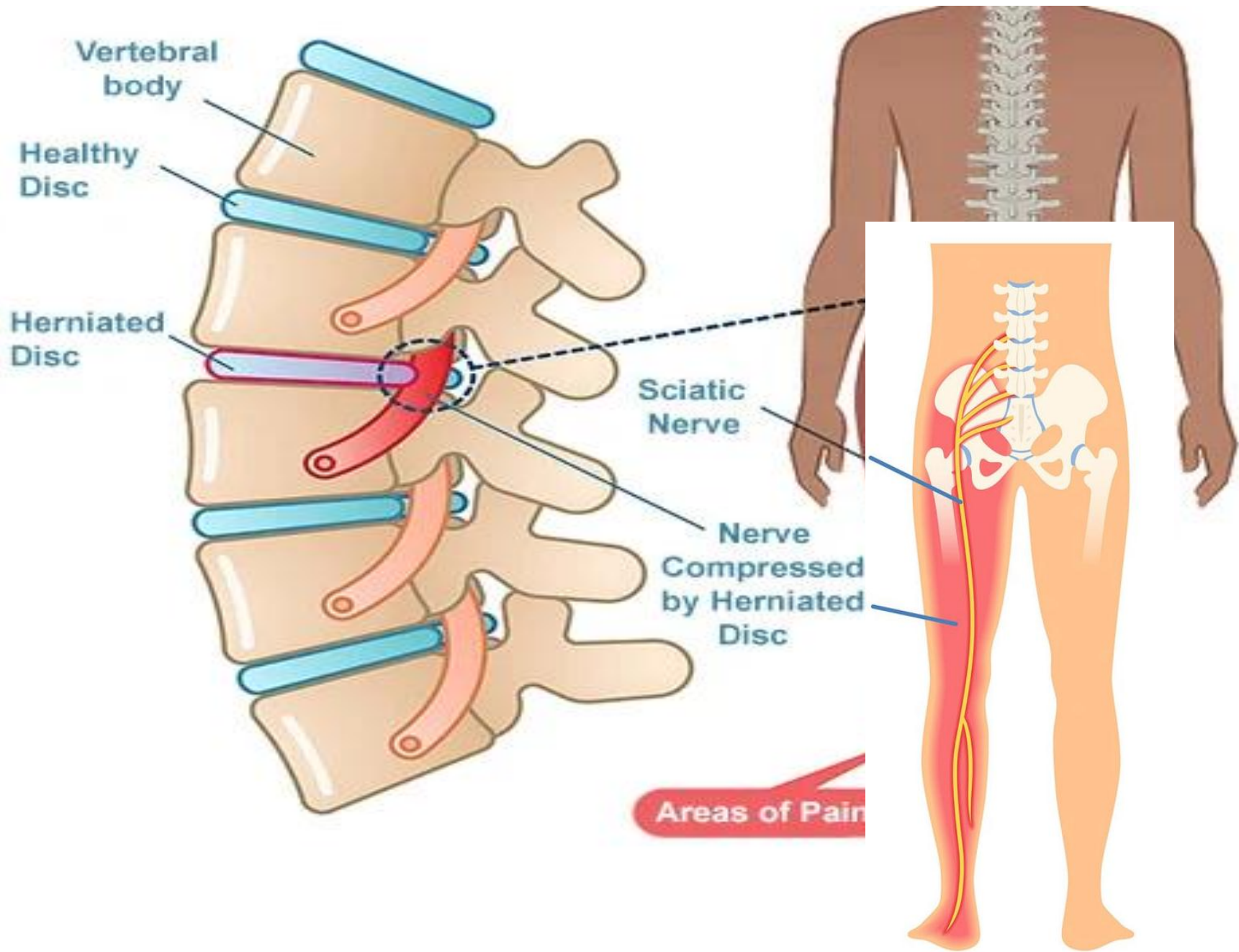
Sciatica (Leg Pain)



Loss of Bladder



Loss of Bladder Control



Assessment and diagnostic studies :

Neurological examination

- Reflexes, muscle strength, sensory deficits
- Straight leg raise test (for lumbar herniation)



COMMONLY TESTED DEEP REFLEXES



Biceps reflex
C5,6



Brachioradialis reflex
C5,6



Triceps reflex
C7,8



Assessment and diagnostic studies :



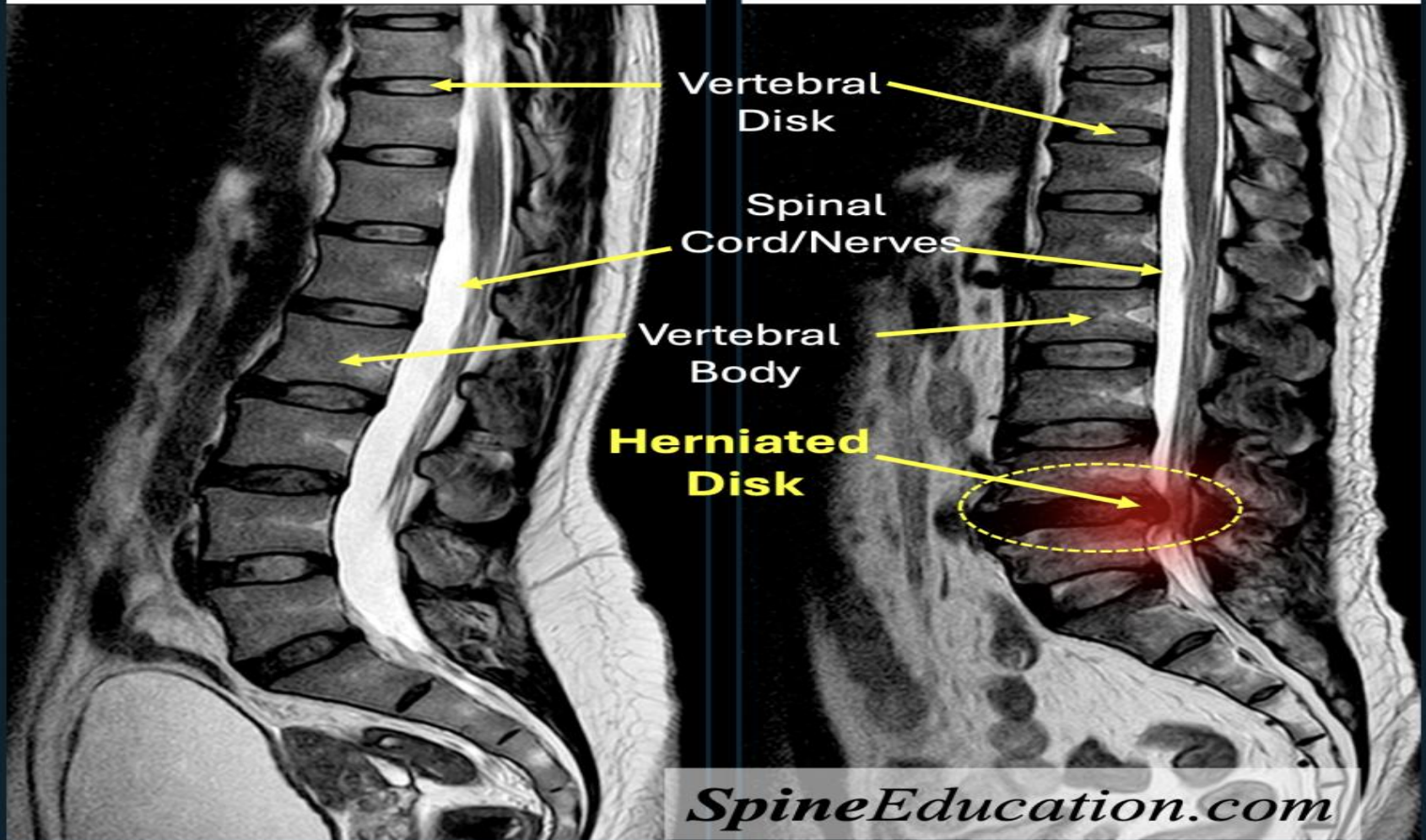
X ray



CT

Normal Lumbar Spine

Herniated Lumbar Disk



MRI

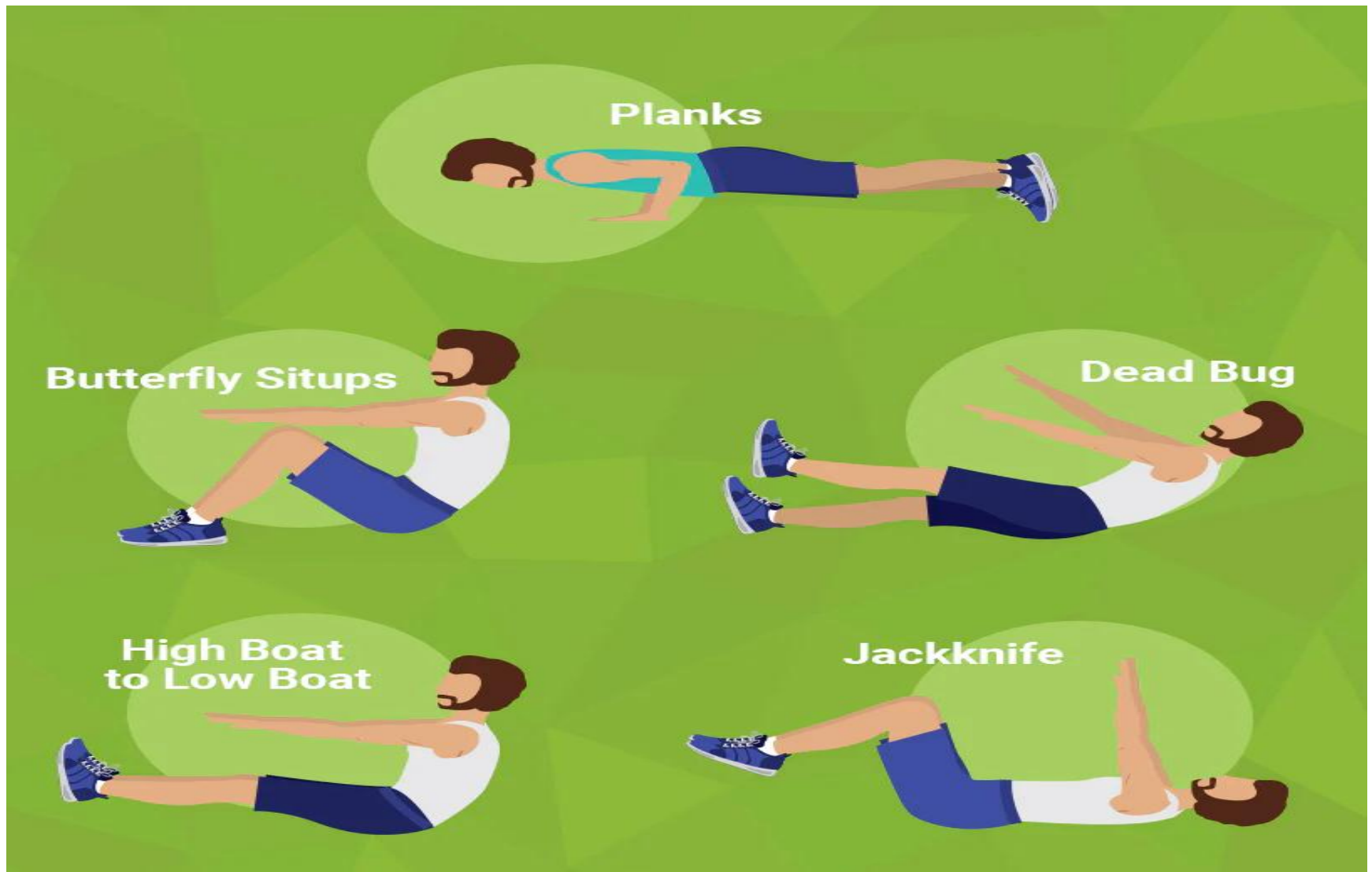
Management

Non-Surgical

- Bed rest (short-term) – avoid prolonged immobility
- NSAIDs (e.g., ibuprofen)
- Muscle relaxants
- Oral steroids or epidural steroid injections
- Hot/cold compresses
- Activity modification (avoid lifting, bending)

• Physiotherapy:

- Core strengthening
- Stretching exercises
- Postural training



Core strengthening



Stretching exercises



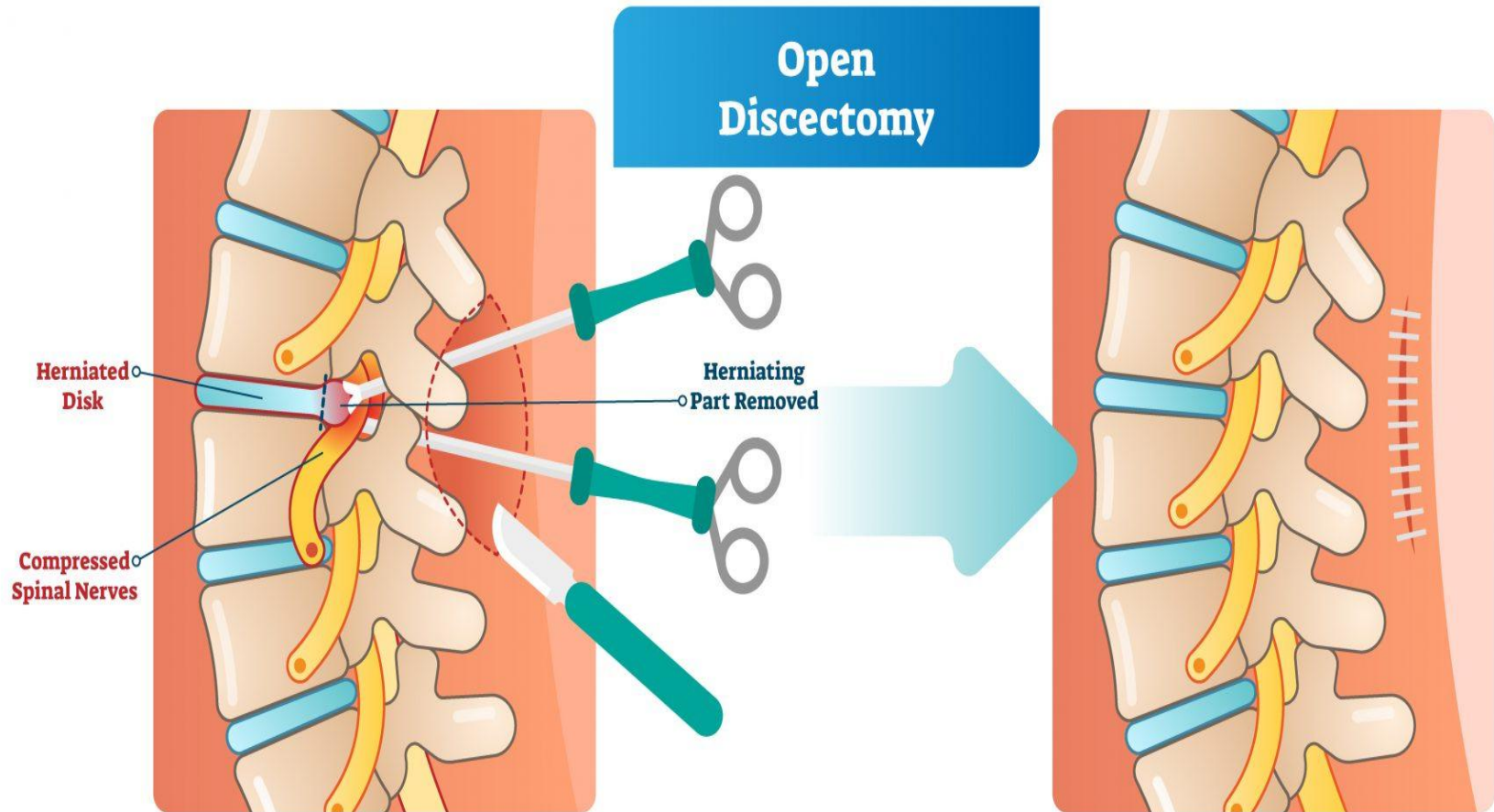
POSTURE CORRECTION



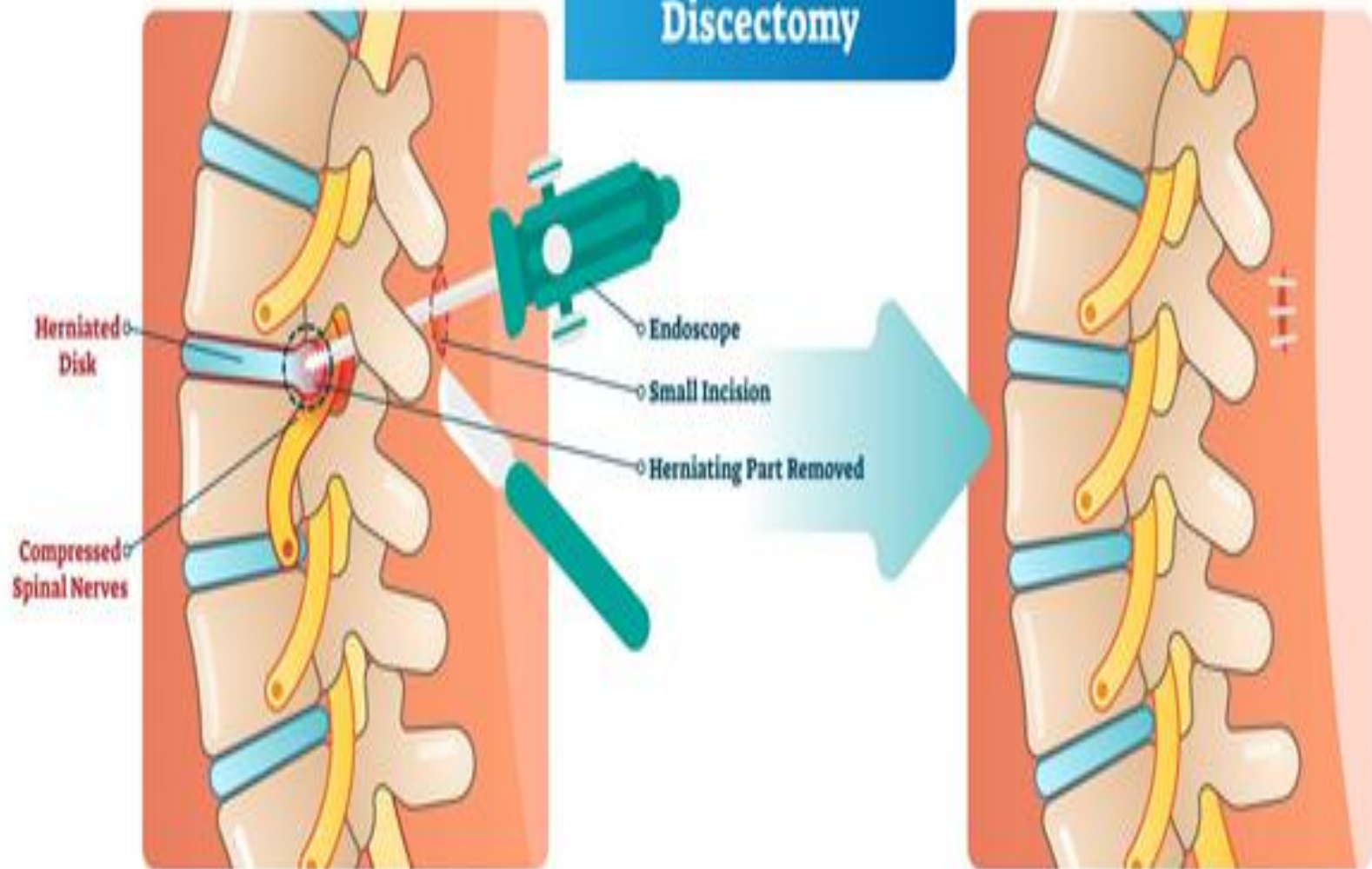
Surgical

- **Discectomy:** removal of the herniated portion of the disc
- **Endoscopic discectomy**
- **Laminectomy:** removal of part of the vertebral bone

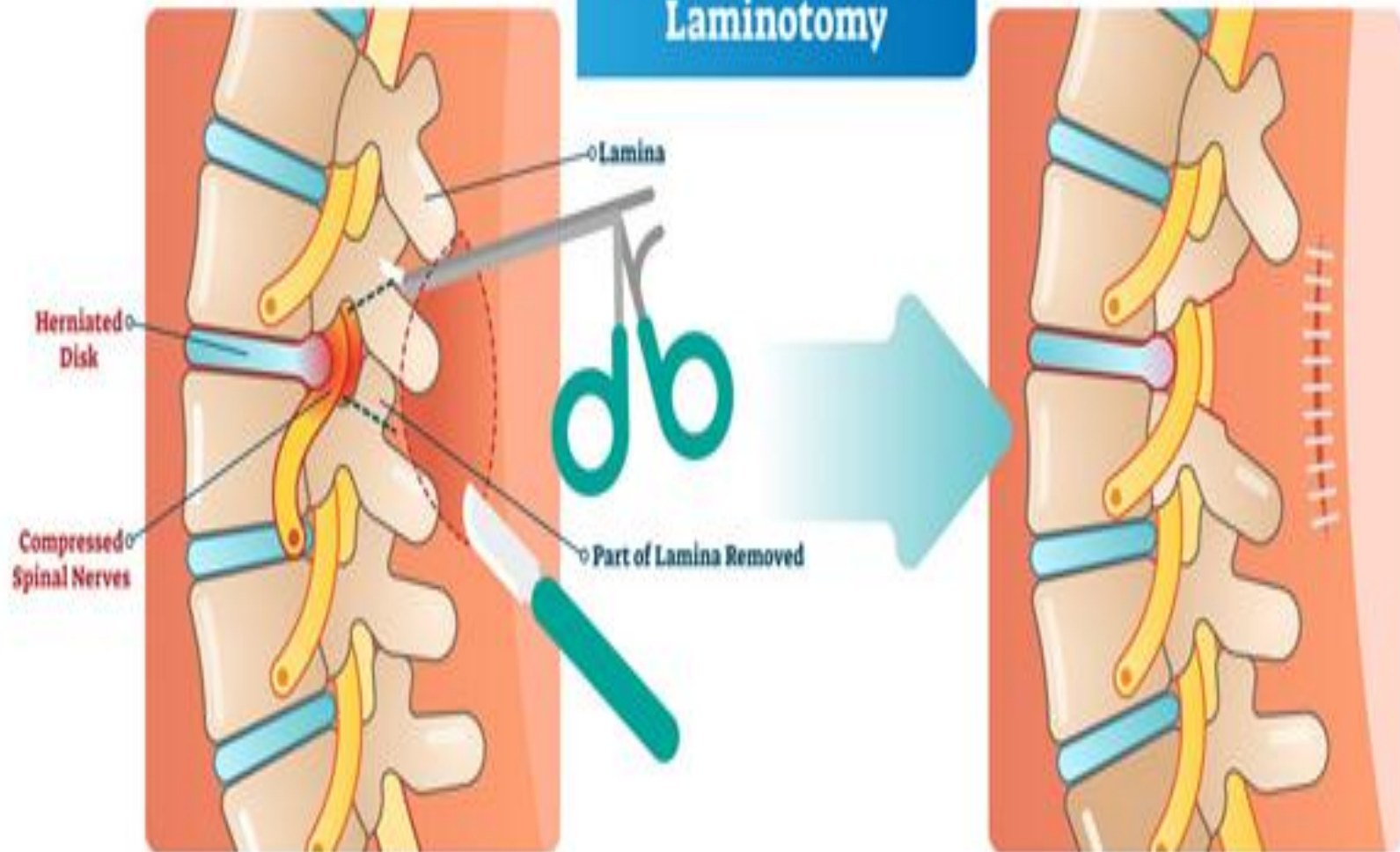
DISC SURGERY



Endoscopic Discectomy



Laminectomy and Laminotomy



Nursing process

1- Assessment :

- Assess **s&s** ; complete assessment of pain
- Observe for numbness, tingling, muscle weakness
- Monitor gait and mobility
- Check for signs of bowel/bladder dysfunction
- Evaluate impact on ADLs (activities of daily living)

Nursing process

2- Nursing diagnoses :

- **Acute Pain** related to nerve compression
- **Impaired physical** mobility related to pain and discomfort
- **Anxiety** related to limited mobility
- **Deficient knowledge** related to condition and self-care
- **Risk for injury** related to muscle weakness or sensory deficits

Nursing process

3,4 - Patient goal & Nursing interventions :

goal : patient will report reduced pain / Pain Management

- Apply hot/cold **compresses**
- Encourage use of **firm mattress** or back support
- Position patient with **knees slightly bent** (lumbar cases)
- Administer **analgesics**, **NSAIDs**, muscle **relaxants** as prescribed

Nursing process

goal : Patient will maintain mobility and function.

Mobility support

- Encourage **mobilization** as tolerated
- Assist with **physical therapy**
- Teach **log-rolling** technique when turning in bed
- Use of **braces** or lumbar support **belts** if prescribed

Nursing process

goals : **Reduce anxiety**

- Allow patient to express feeling.
- Explanation about diagnosis, treatment and medication.
- Good interaction between patient and nurse.
- Encourage the family to participate in patient care and emotional support

Nursing process

Education

- Teach proper **body mechanics**: bend at knees, not waist
- **Avoid heavy lifting**, prolonged sitting, or twisting
- Encourage **weight loss** if overweight
- Emphasize importance of **exercise**, **core strengthening**

5- Evaluation

- Patient reports pain is controlled
- Demonstrates safe mobility
- No signs of neurological deterioration
- Verbalizes understanding of preventive strategies



Summary



- **List 4 manifestations of lumbar disc prolapse.**
- **List 4 non surgical management of disc prolapse**
- **List 3 nursing diagnosis of disc prolapse**



thank
you